Design Brief – Review 2

0. Header

Project code	FA26SE183
Project name	VISTA: Vision-Language-Action System for Tabletop Manipulation with Dual Robot Arms
Team	GFA26SE94
Supervisor	Lâm Hữu Khánh Phương
Repository link (tag review-2)	https://github.com/SKYNET-Thesis/se — tag review-2 currently points to 27c60b1; final documentation not yet tagged
Version	Registration v1.2 — proposed; Review 2 approval pending
Date	27/10/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
Scope	Earlier approved version: not available to compare	Registration v1.2 — four school-to-experiment learning workflows	Review 1 scope alignment; requires verified old version
Robot execution	Hardware E2E proof not supplied	Schematic active in Education API; physical task still target	Must not present planned as implemented

1.2 Out of Scope

Item	Reason
Standalone mobile	Outside mandatory Education Platform Review 2 commitments
Training VLA foundation model from scratch	Outside team SE mandatory scope
Unsupervised cloud control of hardware	Safety boundary of tabletop lab

2. System at a Glance

VISTA is a Robotics & AI Education Platform connecting schools, teachers and students with structured STEM learning and tabletop robot experimentation. School administrators manage classrooms and authorize robot resources; teachers prepare lessons, publish assignments and review student submissions. Students create versioned robot programs, submit work, access permitted experiment activities and receive feedback. The platform uses a React web application, a NestJS Education API, Supabase authentication and

PostgreSQL, with separate AI integration and local robot runtime components. Its core value is traceable, role-scoped learning from assignment to program, experiment evidence and assessment. At Review 2, the Education API supports schematic experiment requests; physical robot execution remains a target requiring end-to-end verification. The architecture and integration boundaries are documented in Report 4 (SDD), C4 System Context and Container diagrams.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
DD-01	Scoped API data access	Direct Supabase browser writes vs guarded NestJS	NestJS Education API + Prisma with membership guard	Lê Nguyễn Quang Đức	FR-01 / C4
DD-02	JWT tenant authorization	RLS-only vs app guards	Supabase JWT and organization/classroom guards	Lê Nguyễn Quang Đức	FR-01 / WF-01
DD-03	Robot entitlement allocation	Live reservation vs school allocation	Atomic classroom authorization separate from runtime	Lê Nguyễn Quang Đức	FR-02 / WF-01
DD-04	Experiment boundary	Coupled hardware calls vs separate runtime	Schematic path + separate local Python runtime	Vũ Anh Tuấn	FR-05 / WF-03
DD-05	Assessment and evidence	Embedded binary vs metadata reference	Store durable refs and teacher assessment separately	Phạm Viết Lợi	FR-06 / WF-04

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R-01	Physical runtime integration unverified	High	Separate schematic request from hardware execution and collect tested E2E evidence	Vũ Anh Tuấn
R-02	Cross-school authorization/data leak	High	Integration tests for JWT/membership/classroom scope	Lê Nguyễn Quang Đức
R-03	Diagram states and ERD constraints may differ from code	High	Validate each state against enum/service and FK/unique	Trần Tiến Đạt

ID	Risk	Impact	How we verify or mitigate	Owner
			constraints	

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G-01	Physical robot end-to-end execution is not verified	Education API / Robot Runtime	Keep schematic and physical claims separate; capture supervised runtime evidence	Vũ Anh Tuấn	Before Defense 1.1
G-02	Design state labels are not mapped to persisted values	Report 4 Figures 08–11	Publish the state-to-code mapping and transition tests	Trần Tiến Đạt	Before official BM02
G-03	Approved registration and Review 1 decision record are unavailable	Registration / BM03	Obtain authentic signed sources and reconcile every scope change	Lê Nguyễn Quang Đức	Before submissio n

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
FR-0 1	Role-scoped access	All roles	All seeded cross-role/cross-org requests are denied without disclosure	TC-FR01-01..03	Implement ed; verify
FR-0 2	Robot allocation	School Admin	Only an entitled robot is allocated; conflict/cross-org requests are denied	TC-FR02-01..03	Implement ed; verify
FR-0 3	Assignment delivery	Teacher	Authorized publish succeeds; unauthorized publish is denied	TC-FR03-01..03	Implement ed; verify
FR-0 4	Versioned submission	Student	Only own eligible immutable version can be submitted	TC-FR04-01..03	Implement ed; verify
FR-0 5	Schematic experiment	Student	Scoped run persists truthful result; invalid mode/access is denied	TC-FR05-01..03	Implement ed; verify
FR-0 6	Physical boundary	Student, Runtime	Eligibility and runtime acknowledgement required; no false success	TC-FR06-01..03	Partial; blocked
FR-0	Assessment/	Teacher,	Scoped	TC-FR07-	Implement

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
7	reflection	Student	assessment/feedback/reflection persists; foreign access denied	01..03	ed; verify
FR-08	AI/data provenance	Student, Teacher	Scoped metadata is traceable; unavailable operation never completes	TC-FR08-01..03	Read-only partial

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
NFR-01	Authorization	Negative scope cases	100% denied	Seeded API tests
NFR-02	Runtime safety	Duplicate/ unavailable/ timeout/stop/loss	No false success	Mock + supervised hardware tests
NFR-03	API latency	P95 ordinary request	≤3 s	Recorded staging load test
NFR-04	Acknowledgement	Accepted request to ack/timeout	≤5 s	Instrumented runtime test
NFR-05	Traceability	Runs with actor/resource/time/state	100%	Seeded evidence query
NFR-06	Automation	Committed tests in CI	≥50%; main 100%; total ≥90%	Tagged test inventory + CI logs

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Lê Nguyễn Quang Đức	Leader	Auth/RBAC, school/robot, CI	WF-01	DD-01–03
Trần Tiến Đạt	Member	Learning/assignments, version/submission, UML	WF-02	Submission states
Vũ Anh Tuấn	Member	Experiment API, robot runtime, safety	WF-03	DD-04
Phạm Viết Lợi	Member	Assessment, feedback, AI metadata	WF-04	DD-05

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	School administration and robot allocation	School Admin, Teacher	Authorized admin allocates an entitled robot	Scoped classroom allocation persisted	FR-01, FR-02	Unallocated → Allocated / Replaced	Lê Nguyễn Quang Đức
WF-02	Learning assignment and submission	Teacher, Student	Teacher publishes classroom work	Eligible learner submits an immutable version	FR-03, FR-04	Assignment Draft → Published; Attempt In Progress → Submitted	Trần Tiến Đạt
WF-03	Experiment execution and evaluation	Student, Robot Runtime	Learner requests an authorized experiment	Truthful schematic result or bounded runtime outcome persisted	FR-05, FR-06	Experiment Requested/Pending → Running → Completed/Failed/Cancelled	Vũ Anh Tuấn
WF-04	Teacher review and learner reflection	Teacher, Student	Teacher opens an eligible submission	Assessment, feedback and reflection persisted	FR-07, FR-08	Submission Submitted → Under Review → Feedback Published	Phạm Viết Lợi

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01a	WF-01	Main	Seeded school, classroom and entitled robot	Open classroom; select eligible robot; allocate	Allocation is persisted and visible to the instructor	Lê Nguyễn Quang Đức
DS-01b	WF-01	Exception	Robot belongs to another organization	Attempt cross-organization allocation	Request is denied and no allocation is written	Lê Nguyễn Quang Đức

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-02a	WF-02	Main	Seeded teacher/student membership and draft assignment	Publish assignment; start attempt; version program; submit	Immutable learner submission is persisted	Trần Tiến Đạt
DS-02b	WF-02	Exception	Learner has no active classroom membership	Open or submit foreign classroom work	Request is denied without record disclosure	Trần Tiến Đạt
DS-03a	WF-03	Main	Eligible seeded attempt and valid schematic program	Create experiment; run schematic; evaluate result	Truthful result, events and evidence are persisted	Vũ Anh Tuấn
DS-03b	WF-03	Exception	Unsupported/ unsafe physical request or unavailable runtime	Request physical execution	Request is rejected/stopped and never marked successful	Vũ Anh Tuấn
DS-04a	WF-04	Main	Submitted work with seeded rubric and authorized teacher	Open evidence; assess; publish feedback; learner reflects	Assessment, feedback and reflection are persisted and scoped	Phạm Viết Lợi
DS-04b	WF-04	Exception	Teacher or learner belongs to another classroom	Attempt review or feedback access	Request is denied without exposing the submission	Phạm Viết Lợi

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	8	3 per feature	24
Main workflows	4	1 per workflow	4
State transitions	48	1 per enumerated valid/invalid transition	48
NFRs	6	1 measurement per NFR	6
TOTAL PROPOSED	82	Review 2 approval	82